

APP DEV 2 PROJECT REPORT

1. Student Details

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About Me: I am a diploma level student in the IITM BS DEGREE in Data Science and Applications.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build an efficient to manage campus recruitment activities involving companies, students, and placement drives.

Approach:

The app was built using Flask as the backend framework and Vue3 as the frontend framework. It allows companies to create placement drives and manage applications and students to create their profiles, apply for placement drives and track their application status.

3. AI/LLM Declaration

I used ChatGPT (GPT-5) to assist in writing SQLAlchemy model definitions,improving ui design (bootstrap).

The extent of AI/LLM usage is around **15-20%**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Flask-RESTful	Flask extension for building REST API
Bootstrap 5	Frontend styling and responsive design
Flask-Security	User and role management and token authentication
SQLite	Lightweight local database for storing user data
Vue3	Frontend framework
Redis	Message broker
Celery	Distributed task queue used to run background jobs
Pinia	State management framework for Vue3
Flask-Caching	Caches API responses

5. Video Presentation

<https://drive.google.com/file/d/17sPvrBbUULZsIDHGNOy7fQ1u0ypRBH5o/view?usp=sharing>

6. Database Schema / ER Diagram

Tables

User — stores authentication and account information

(id, username, email, password, active, fs_uniquifier, fs_token_uniquifier, created_at)

Role — stores system roles

(id, name, description)

UserRoles — mapping table for assigning roles to users
(user_id, role_id)

Student — stores student profile information
(id, user_id, name, roll_number, department, cgpa, resume, passing_out_year)

Company — stores company profile information
(id, user_id, company_name, hr_contact, website, approval_status)

PlacementDrive — stores job/placement drives created by companies
(id, company_id, title, description, min_cgpa, passing_out_year, application_deadline, status, created_at)

Application — stores student applications to placement drives
(id, student_id, drive_id, applied_on, status)

Relationships

Many-to-Many → User → Role

- Implemented through **UserRoles** table.

One-to-One → User → Student

- Each user with student role has one student profile.

One-to-One → User → Company

- Each user with company role has one company profile.

One-to-Many → Company → PlacementDrive

- One company can create multiple placement drives.

One-to-Many → Student → Application

- One student can apply to multiple placement drives.

One-to-Many → PlacementDrive → Application

- One placement drive can receive multiple applications.
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roles		
int	id	PK
string	name	
string	description	

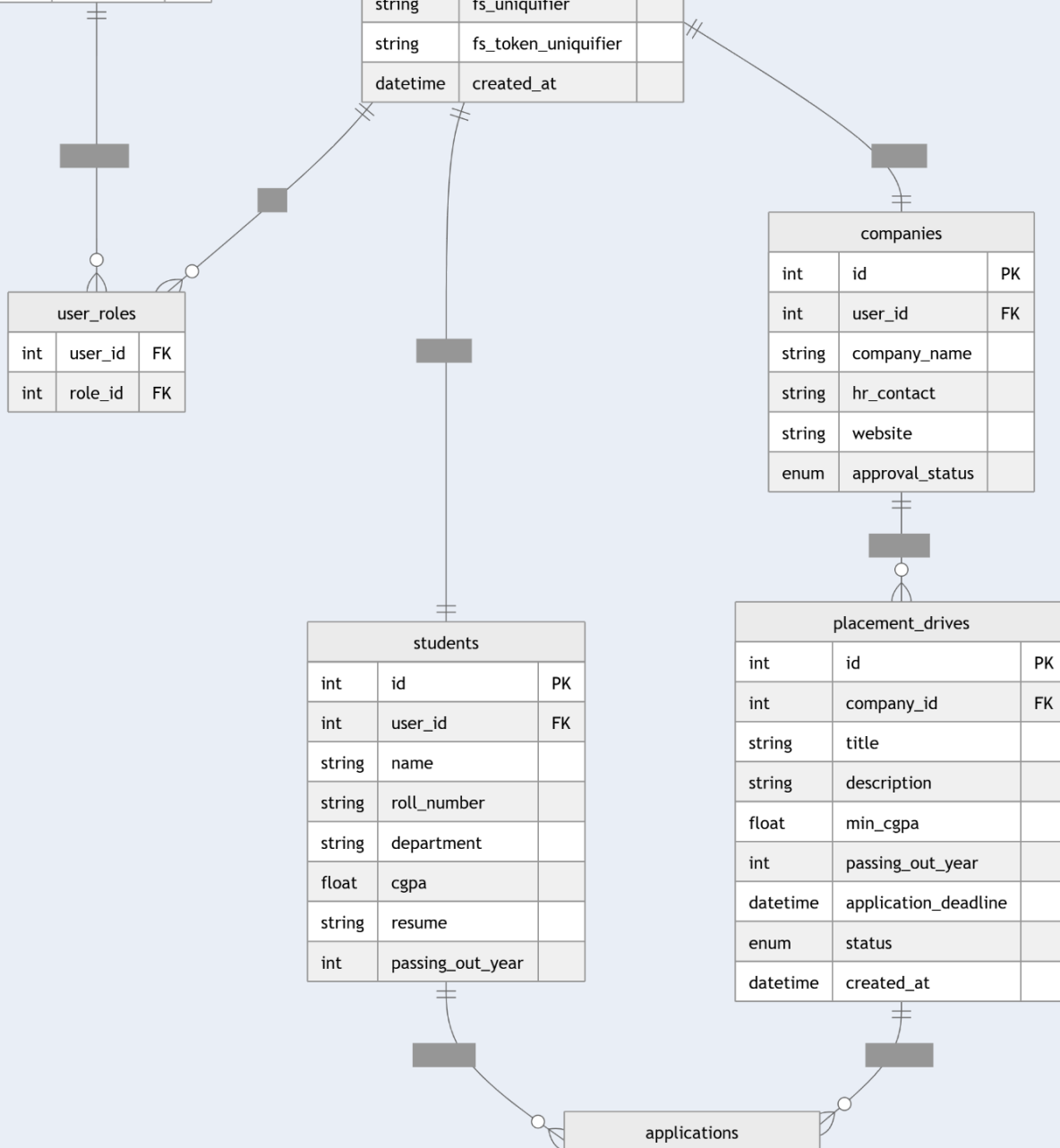
users		
int	id	PK
string	username	
string	email	
string	password	
boolean	active	
string	fs_uniquifier	
string	fs_token_uniquifier	
datetime	created_at	

companies		
int	id	PK
int	user_id	FK
string	company_name	
string	hr_contact	
string	website	
enum	approval_status	

students		
int	id	PK
int	user_id	FK
string	name	
string	roll_number	
string	department	
float	cgpa	
string	resume	
int	passing_out_year	

placement_drives		
int	id	PK
int	company_id	FK
string	title	
string	description	
float	min_cgpa	
int	passing_out_year	
datetime	application_deadline	
enum	status	
datetime	created_at	

applications		
int	id	PK
int	student_id	FK
int	drive_id	FK
datetime	applied_on	
enum	status	



7. API Resource Endpoints

Authentication APIs

Endpoint	Method	Description
/api/login	POST	Authenticate user and generate authentication token
/api/logout	POST	Logout user and invalidate authentication token
/api/register	POST	Register a new user (student or company)

Common APIs

Endpoint	Method	Description
/api/company/<company_id>	GET	Retrieve company details
/api/admin/company/<company_id>	GET	Admin view of company details
/api/company/drives/<drive_id>	GET	Retrieve details of a specific placement drive
/api/admin/drive/<drive_id>	GET	Admin view of placement drive details
/api/student/<student_id>	GET	Retrieve student profile details
/api/admin/student/<student_id>	GET	Admin view of student profile
/api/resume/	GET	View or download student resume

Company APIs

Endpoint	Method	Description
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/api/company/drives	GET	Retrieve placement drives created by the company
/api/company/create_drive	POST	Create a new placement drive
/api/company/drives/<drive_id>/close	PUT	Close an existing placement drive
/api/company/drive/<drive_id>/applications	GET	Retrieve all applications for a specific drive
/api/company/application/<application_id>	GET	Retrieve details of a specific application
/api/company/application/change_status/<application_id>	PUT	Update application status (shortlist, select, reject)

Student APIs

Endpoint	Method	Description
/api/student/approved_drives	GET	Retrieve list of approved placement drives
/api/student/drive/<drive_id>	GET	Retrieve placement drive details
/api/student/drive/<drive_id>/apply	POST	Apply to a placement drive
/api/student/edit_profile	PUT	Update student profile information
/api/student/applications	GET	Retrieve student application history
/api/student/upload_resume	POST	Upload or update student resume

Admin APIs

Endpoint	Method	Description
/api/admin/dashboard	GET	Retrieve admin dashboard statistics
/api/admin/drive/<drive_id>/approval	PUT	Approve or reject placement drive
/api/admin/company/registrations	GET	Retrieve pending company registrations
/api/admin/company/approval/<company_id>	PUT	Approve or reject company registration
/api/admin/companies	GET	Retrieve list of all companies
/api/admin/company/blacklist/<role_based_id>	PUT	Blacklist or unblacklist a company
/api/admin/student/blacklist/<role_based_id>	PUT	Blacklist or unblacklist a student
/api/admin/students	GET	Retrieve list of all students
/api/admin/drives	GET	Retrieve list of all placement drives
/api/admin/drive/<drive_id>/applications	GET	Retrieve all applications for a specific drive
